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SUMMER INTERNSHIP REPORT

PROJECT TITLE

AI-Driven HY Kero Flash Point Prediction and Process Optimization

Submitted in partial fulfillment of the Summer Internship Programme

Organization

Indian Oil Corporation Limited (IOCL)

Mathura Refinery

Information Systems Department

Internship Duration

15 June 2026 – 14 July 2026

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CERTIFICATE

This is to certify that the Summer Internship Project entitled "AI-Driven HY Kero Flash Point Prediction and Process Optimization" has been successfully carried out at Indian Oil Corporation Limited (IOCL), Mathura Refinery, Information Systems Department during the period 15 June 2026 to 14 July 2026 by:

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The project was completed under the guidance of Mr. S. K. Yadav (Project Guide) and the mentorship of Mr. S. K. Pandey (Project Mentor).

The work presented in this report is original and demonstrates the application of Artificial Intelligence and Machine Learning techniques for real-time Heavy Kerosene Flash Point prediction and refinery process optimization.

This report is submitted in partial fulfillment of the Summer Internship Programme.

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We sincerely thank all the officers and staff members of the Information Systems Department for their cooperation, technical assistance, and willingness to share their valuable knowledge and industrial experience.

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Finally, we would like to thank our team members for their dedication, cooperation, teamwork, and continuous support during the successful completion of this internship project.

Team Members

- **Gun Sharma**
- **Divyanshi Upman**
- **Aryan Anand**
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ABSTRACT

The petroleum refining industry continuously seeks innovative technologies to improve product quality, operational efficiency, and process safety. One of the critical quality parameters of Heavy Kerosene (HY Kero) is its Flash Point, which determines the safe handling, storage, and transportation of petroleum products. Traditionally, the flash point is determined through laboratory testing, resulting in delayed feedback for refinery operators.

This internship project, titled "AI-Driven HY Kero Flash Point Prediction and Process Optimization," aims to develop an Artificial Intelligence-based predictive system capable of estimating the Heavy Kerosene Flash Point using historical refinery process data. The project was carried out at the Information Systems Department, Indian Oil Corporation Limited (IOCL), Mathura Refinery.

Historical process variables such as temperatures, pressures, flow rates, furnace outlet temperatures, and stripping steam parameters were collected and analyzed. Data preprocessing, feature engineering, and machine learning techniques were applied to develop predictive models. Several regression algorithms including Linear Regression, Ridge Regression, Lasso Regression, ElasticNet, Random Forest, XGBoost, LightGBM, Bayesian Ridge, and Huber Regression were evaluated.

Among all the models, the Huber Tuned Regression Model demonstrated the best performance with high prediction accuracy and robustness against industrial data noise. A web-based application was also developed using FastAPI, React, and SQLite, providing features such as real-time prediction, prediction history, model comparison, dashboard visualization, and batch prediction.

The developed system enables continuous monitoring of Heavy Kerosene Flash Point, reduces dependence on laboratory testing, supports timely operational decisions, improves product quality, and contributes to refinery process optimization. The project demonstrates the practical application of Artificial Intelligence and Machine Learning in modern petroleum refining industries.

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CHAPTER 1

INTRODUCTION

1.1 Introduction

The petroleum refining industry plays a vital role in fulfilling the energy requirements of transportation, aviation, industries, and households. As refinery operations become increasingly complex, maintaining product quality while maximizing operational efficiency has become a major challenge. Modern refineries generate thousands of process measurements every minute through Distributed Control Systems (DCS), creating opportunities for applying Artificial Intelligence (AI) and Machine Learning (ML) to improve process monitoring and decision-making.

One of the most important quality parameters of Heavy Kerosene (HY Kero) is its Flash Point, which indicates the minimum temperature at which kerosene vapours ignite when exposed to an ignition source. Maintaining the flash point within the prescribed specification is essential to ensure product safety, transportation compliance, and customer satisfaction.

Traditionally, the flash point is measured through laboratory testing using Gas Chromatography (GC). Although laboratory analysis provides accurate results, it requires considerable time to collect samples, perform testing, and communicate the results to plant operators. During this delay, the operating conditions of the refinery may change significantly, making corrective actions reactive rather than proactive.

Artificial Intelligence provides an effective solution to this problem by enabling the development of Soft Sensors. A Soft Sensor is a data-driven predictive model that estimates laboratory quality parameters using continuously available process variables. Instead of waiting for laboratory results, operators can obtain instant predictions and take corrective actions whenever required.

During our internship at Indian Oil Corporation Limited (IOCL), Mathura Refinery, Information Systems Department, we developed an AI-based system capable of predicting the Heavy Kerosene Flash Point in real time. The system utilizes historical refinery data, applies data preprocessing and feature engineering techniques, and trains multiple machine learning algorithms to identify the most accurate predictive model.

The developed application also includes a web-based dashboard for real-time prediction, prediction history, batch prediction, model comparison, and data visualization. This project demonstrates how Artificial Intelligence can support refinery operations by reducing laboratory dependency, improving operational efficiency, minimizing energy consumption, and ensuring consistent product quality.

1.2 Need of the Project

In conventional refinery operations, laboratory testing introduces delays in quality monitoring, making it difficult for operators to respond quickly to changing process conditions. Real-time prediction of Heavy Kerosene Flash Point can significantly improve process control by providing immediate feedback based on live process data.

The implementation of an AI-driven prediction system reduces dependence on laboratory analysis, minimizes production losses, improves operational efficiency, and supports intelligent decision-making. Such systems represent an important step towards Industry 4.0 and digital transformation in petroleum refineries.

1.3 Objectives of the Project

The major objectives of this internship project are:

- To understand the Heavy Kerosene production process in the Crude Distillation Unit.
- To study the importance of Flash Point in refinery product quality.
- To analyze historical refinery process data.
- To preprocess industrial datasets for machine learning.
- To develop an AI-based Flash Point prediction model.
- To compare different machine learning algorithms.
- To deploy the best-performing model in a web application.
- To assist refinery engineers in real-time decision-making.
- To reduce dependence on laboratory testing.
- To improve refinery process optimization.

1.4 Scope of the Project

The project focuses on developing a machine learning model capable of predicting Heavy Kerosene Flash Point using historical process data collected from refinery sensors. The developed system is intended to support refinery engineers by providing real-time predictions through an interactive dashboard.

The project includes data preprocessing, feature engineering, model training, evaluation, visualization, and deployment. Although the developed system is designed for Heavy Kerosene Flash Point prediction, the same methodology can be extended to other refinery quality parameters such as diesel cetane number, gasoline octane number, sulphur content, and distillation characteristics.

1.5 Organization of the Report

This report is organized into multiple chapters covering various aspects of the internship project. It begins with an introduction to the project, followed by details of IOCL and Mathura Refinery, the Information Systems Department, project objectives, literature review, methodology, system development, implementation, results, conclusion, and future scope.

CHAPTER 2

ABOUT IOCL AND MATHURA REFINERY

2.1 About Indian Oil Corporation Limited (IOCL)

Indian Oil Corporation Limited (IOCL) is India's largest integrated energy company and a Fortune Global 500 enterprise. Established in 1959, the company operates under the ownership of the Government of India and plays a significant role in meeting the country's energy demands. IOCL is engaged in refining crude oil, transporting petroleum products through pipelines, marketing fuels, manufacturing petrochemicals, and promoting renewable energy initiatives.

The corporation operates an extensive network of refineries, pipelines, terminals, research centres, and marketing divisions across India. With a strong commitment to innovation and sustainability, IOCL has adopted advanced digital technologies such as Artificial Intelligence, Industrial Internet of Things (IIoT), Big Data Analytics, and Predictive Maintenance to enhance operational efficiency and environmental performance.

The company's vision is to become "A globally admired company delivering energy solutions through sustainable practices, innovation, and operational excellence."

2.2 About Mathura Refinery

Mathura Refinery, located in Uttar Pradesh, is one of the major refineries of Indian Oil Corporation Limited. Commissioned in 1982, it has significantly contributed to the petroleum requirements of northern India by producing high-quality petroleum products.

The refinery processes various grades of crude oil and produces Liquefied Petroleum Gas (LPG), Motor Spirit (Petrol), High-Speed Diesel (HSD), Aviation Turbine Fuel (ATF), Superior Kerosene Oil (SKO), Heavy Kerosene (HY Kero), Naphtha, Bitumen, and other petroleum products.

The refinery consists of several major processing units, including the Crude Distillation Unit (CDU), Vacuum Distillation Unit (VDU), Diesel Hydro Desulphurization Unit (DHDS), Catalytic Reforming Unit (CRU), and Sulphur Recovery Unit (SRU). These units work together to convert crude oil into valuable petroleum products while maintaining high standards of safety, quality, and environmental compliance.

Mathura Refinery has embraced digital transformation by implementing modern technologies such as Artificial Intelligence, Machine Learning, Industrial Automation,

and Data Analytics. These initiatives help improve product quality, reduce operational costs, optimize energy consumption, and support predictive maintenance.

The internship conducted at the Information Systems Department provided practical exposure to these digital technologies and their application in refinery process optimization. Through this project, we gained valuable insights into the integration of AI with refinery operations, particularly for real-time Heavy Kerosene Flash Point prediction.

CHAPTER 3

INFORMATION SYSTEMS (IS) DEPARTMENT

3.1 Introduction

The Information Systems (IS) Department is one of the key departments at IOCL Mathura Refinery responsible for managing the refinery's digital infrastructure and information technology systems. The department ensures the smooth functioning of computer networks, enterprise applications, databases, cybersecurity, industrial communication systems, and digital transformation initiatives across the refinery.

With the rapid advancement of Industry 4.0 technologies, the IS Department plays a significant role in integrating Artificial Intelligence (AI), Machine Learning (ML), Data Analytics, and Industrial Internet of Things (IIoT) into refinery operations. These technologies enable better monitoring, predictive analysis, process optimization, and informed decision-making.

During our internship, the IS Department provided technical guidance, historical refinery process data, and practical exposure to industrial software applications. This support enabled us to develop an AI-driven solution for predicting Heavy Kerosene Flash Point and understanding how machine learning can be applied in real industrial environments.

3.2 Functions of the IS Department

The major responsibilities of the Information Systems Department include:

- Management of refinery IT infrastructure.
- Administration of industrial databases and servers.
- Maintenance of Distributed Control System (DCS) data.
- Network administration and cybersecurity.
- Development of software applications for refinery operations.
- Data visualization and reporting.
- Backup and disaster recovery management.
- Support for enterprise applications.
- Implementation of Artificial Intelligence and Machine Learning projects.
- Digital transformation initiatives across refinery operations.

3.3 Technologies Used by the IS Department

The IS Department utilizes various modern technologies for efficient refinery operations.

Software Technologies

- Database Management Systems
- ERP Applications
- Industrial Data Historians
- SCADA Monitoring Systems
- DCS Integration Tools
- Microsoft Power BI
- Python
- SQL
- Web Technologies

Hardware Infrastructure

- High-performance servers
- Industrial networking equipment
- Storage systems
- Firewall and security appliances
- Backup servers
- Operator workstations

3.4 Learning During the Internship

During the internship, we gained practical knowledge in several areas, including:

- Understanding refinery operations.
- Industrial process data acquisition.
- Process variable analysis.
- Data preprocessing techniques.

- Machine Learning model development.
- Dashboard development.
- Data visualization.
- Model deployment.
- Team collaboration.
- Industrial project documentation.

The internship provided valuable industrial exposure and helped bridge the gap between theoretical knowledge and real-world applications.

CHAPTER 4

PROJECT OVERVIEW

4.1 Project Title

AI-Driven HY Kero Flash Point Prediction and Process Optimization

4.2 Introduction to the Project

Heavy Kerosene (HY Kero) is one of the important petroleum products obtained from the Crude Distillation Unit (CDU). The quality of Heavy Kerosene is determined using several laboratory tests, among which Flash Point is one of the most critical.

Conventionally, flash point is measured through laboratory Gas Chromatography (GC), resulting in delayed feedback to refinery operators. Since laboratory analysis is performed only at specific intervals, operators are unable to monitor product quality continuously.

To overcome this limitation, an Artificial Intelligence-based predictive system was developed during this internship. The system estimates Heavy Kerosene Flash Point using refinery process parameters collected from the Distributed Control System (DCS).

Instead of waiting for laboratory results, operators can obtain instant predictions using the developed web application.

4.3 Problem Identification

The existing laboratory-based approach has several operational limitations:

- Delayed laboratory analysis.
- Lack of continuous quality monitoring.
- Increased dependence on manual testing.
- Slow operator response to process variations.
- Product quality fluctuations.
- Increased operational costs.
- Energy inefficiency.
- Possibility of producing off-specification products.

These challenges motivated the development of an AI-based predictive solution.

4.4 Project Objectives

The project aims to:

- Predict Heavy Kerosene Flash Point in real time.
- Reduce dependence on laboratory testing.
- Improve refinery operational efficiency.
- Enable proactive decision-making.
- Minimize product quality variations.
- Improve process optimization.
- Develop an industrial web application.
- Evaluate multiple machine learning models.
- Deploy the most accurate prediction model.

4.5 Project Methodology

The project followed a systematic methodology consisting of the following stages:

Step 1: Collection of historical refinery process data.

↓

Step 2: Data cleaning and preprocessing.

↓

Step 3: Feature engineering and feature selection.

↓

Step 4: Machine Learning model training.

↓

Step 5: Model evaluation and comparison.

↓

Step 6: Selection of the best-performing model.

↓

Step 7: Web application development.

↓

Step 8: Deployment and testing.

4.6 Software and Tools Used

The following software and technologies were used during the project:

Software/Tool Purpose

Python	Machine Learning
FastAPI	Backend Development
React.js	Frontend Development
SQLite	Database
Pandas	Data Analysis
NumPy	Numerical Computation
Scikit-learn	Machine Learning
Matplotlib	Data Visualization
Joblib	Model Serialization
VS Code	Development Environment
Git	Version Control

4.7 Machine Learning Models Evaluated

Several regression algorithms were trained and evaluated to identify the best-performing model:

- Linear Regression
- Ridge Regression
- Lasso Regression
- ElasticNet Regression
- Bayesian Ridge Regression
- Random Forest Regressor

- XGBoost Regressor
- LightGBM Regressor
- Ensemble Regression
- **Huber Tuned Regression (Selected Model)**

The Huber Tuned Regression model demonstrated the highest prediction accuracy and robustness for refinery data and was therefore selected for deployment.

4.8 Expected Benefits

The developed AI system offers several advantages:

- Continuous Flash Point prediction.
- Faster operational decisions.
- Improved refinery efficiency.
- Reduced laboratory workload.
- Better product quality.
- Lower energy consumption.
- Improved process safety.
- Historical prediction storage.
- Data-driven process optimization.
- Support for Industry 4.0 initiatives.

CHAPTER 5

LITERATURE REVIEW

5.1 Introduction

The rapid advancement of Artificial Intelligence (AI) and Machine Learning (ML) has transformed the way industrial processes are monitored and optimized. Modern petroleum refineries generate large volumes of operational data through sensors installed across different processing units. This data can be utilized to develop predictive models capable of estimating important quality parameters without relying entirely on laboratory testing.

One of the most promising applications of AI in petroleum refining is the development of Soft Sensors. Soft Sensors are mathematical or machine learning models that estimate difficult-to-measure process variables using readily available plant measurements. They provide continuous predictions, enabling engineers to make timely operational decisions and improve process efficiency.

The Heavy Kerosene Flash Point is one such parameter where AI-based prediction systems can significantly reduce delays associated with laboratory analysis while maintaining product quality.

5.2 Artificial Intelligence in Petroleum Refineries

Artificial Intelligence has become an essential technology in modern refineries due to its ability to analyze complex industrial datasets and discover hidden relationships among process variables.

Some common applications include:

- Predictive maintenance
- Fault detection and diagnosis
- Process optimization
- Product quality prediction
- Energy optimization
- Inventory forecasting
- Production planning
- Safety monitoring

AI techniques help refinery engineers improve productivity while reducing operational costs and equipment failures.

5.3 Machine Learning in Process Industries

Machine Learning enables computers to learn patterns from historical data without explicit programming. In petroleum refineries, ML models can accurately estimate product properties using continuously available process variables.

The general machine learning workflow consists of:

- Data Collection
- Data Cleaning
- Feature Engineering
- Model Training
- Model Evaluation
- Deployment
- Continuous Monitoring

This workflow forms the foundation of the AI-driven Flash Point Prediction System developed during the internship.

5.4 Soft Sensor Technology

Soft Sensors are virtual sensors that estimate laboratory measurements using mathematical or machine learning models.

Unlike physical laboratory instruments, Soft Sensors:

- operate continuously,
- provide instant predictions,
- require no manual sampling,
- reduce laboratory workload,
- improve process monitoring.

In petroleum refining, Soft Sensors are widely used for predicting:

- Flash Point

- Octane Number
- Cetane Number
- Sulphur Content
- Density
- Distillation Characteristics
- Product Yield

The present project develops a Soft Sensor for predicting Heavy Kerosene Flash Point.

5.5 Previous Research

Several researchers have proposed AI-based quality prediction systems for petroleum refineries.

Linear Regression models provide simple and interpretable predictions but often fail to capture nonlinear relationships. Random Forest and Gradient Boosting algorithms improve prediction accuracy but may suffer from overfitting when process conditions change.

Recent studies indicate that Huber Regression performs well in industrial environments because it is less affected by noisy laboratory measurements and outliers. This motivated the selection of Huber Regression for the present project.

5.6 Summary

The literature survey indicates that Artificial Intelligence has become a reliable solution for industrial quality prediction. Soft Sensor technology has demonstrated significant benefits in reducing laboratory dependency and improving operational efficiency.

Based on these findings, the present project develops an AI-driven Flash Point Prediction System specifically designed for Heavy Kerosene at IOCL Mathura Refinery.

CHAPTER 6

EXISTING SYSTEM

6.1 Current Method

At present, Heavy Kerosene Flash Point is measured using laboratory testing. Samples are collected from the Crude Distillation Unit and sent to the Quality Control Laboratory, where Flash Point is determined using standard laboratory procedures.

The process involves:

1. Sample Collection
2. Laboratory Testing
3. Result Verification
4. Operator Notification
5. Process Adjustment

Although laboratory testing is highly accurate, it introduces delays between actual process conditions and operator response.

6.2 Limitations of Existing System

The current laboratory-based approach suffers from several limitations:

- Delayed feedback.
- No real-time monitoring.
- High laboratory workload.
- Manual sampling.
- Higher operational cost.
- Delayed corrective actions.
- Increased possibility of off-specification products.
- Inefficient utilization of process data.

These limitations reduce the overall efficiency of refinery operations.

6.3 Need for Automation

Continuous prediction of Heavy Kerosene Flash Point enables refinery operators to respond immediately to process variations. Automation reduces manual effort, improves productivity, and enhances product quality.

An AI-based prediction system provides:

- Continuous monitoring.
- Instant predictions.
- Reduced laboratory dependence.
- Better process optimization.
- Improved decision-making.

CHAPTER 7

PROPOSED AI-BASED SOLUTION

7.1 Proposed System

To overcome the limitations of the conventional laboratory approach, an AI-driven prediction system was developed.

The proposed system uses historical process data collected from the Distributed Control System (DCS). These data are processed and analyzed using Machine Learning algorithms to estimate Heavy Kerosene Flash Point.

The developed application predicts Flash Point within seconds after receiving process inputs.

7.2 System Architecture

The proposed system consists of five major components:

1. Data Collection Layer

Historical refinery process data collected from DCS sensors.

↓

2. Data Processing Layer

- Data Cleaning
- Missing Value Handling
- Feature Engineering
- Data Normalization

↓

3. Machine Learning Layer

Training multiple regression algorithms.

↓

4. Prediction Layer

Best-performing model (Huber Regression) generates Flash Point predictions.

↓

5. User Interface

Interactive web application for operators.

7.3 Functional Modules

The application consists of the following modules:

Dashboard

Displays

- Overall statistics
- Prediction summary
- Model information
- Historical graphs



Prediction Module

Allows users to enter process values manually and receive Flash Point predictions instantly.

HY KERO

FLASH POINT PREDICTOR

PREDICT

HY Kero Flash Point Prediction - Crude Distillation Unit

MODEL ACTIVE (HUBER TUNED)

Dashboard

Predict

History

Models

Upload

FLASH POINT PREDICTOR

Enter process conditions to get an instant Flash GC estimate with confidence bounds

MANUAL INPUT TIME WINDOW

Try a scenario: use the quick presets below to explore normal, watch, and high-risk operating conditions.

NORMAL RUN

Stable conditions near expected operating band

WATCHLIST

Mild drift with rising furnace duty

HIGH RISK

Potential off-spec trend with elevated severity

PROCESS SENSOR VALUES (±30 MIN WINDOW WEAR)

HK DRAB TEMPERATURE (°C)

224

FLASH ZONE TEMPERATURE (°C)

365

COLUMN TOP TEMPERATURE (°C)

151

FURNACE 15F1 OUTLET TEMP (°C)

368

FURNACE 15F2 OUTLET TEMP (°C)

367

FURNACE 15F3 OUTLET TEMP (°C)

369

FURNACE 15F4 OUTLET TEMP (°C)

368

HK DRAB FLOW RATE (M³/HR)

95

STRIPPING STEAM SIZES (TPH)

5.0

LAB-1 FLASH GC (°C) - PREVIOUS SHIFT

64.38

LAB-2 FLASH GC (°C) - 8 SHIFTS AGO

64.38

PREDICT

RESET

PREDICTED FLASH GC

64.4

DEGREES CELSIUS

95% CI: [66.5°C - 66.2°C]

IN SPEC

In Spec

Current conditions look within expected operating bounds.

PREDICTION DETAILS

PREDICTION COMPLETED

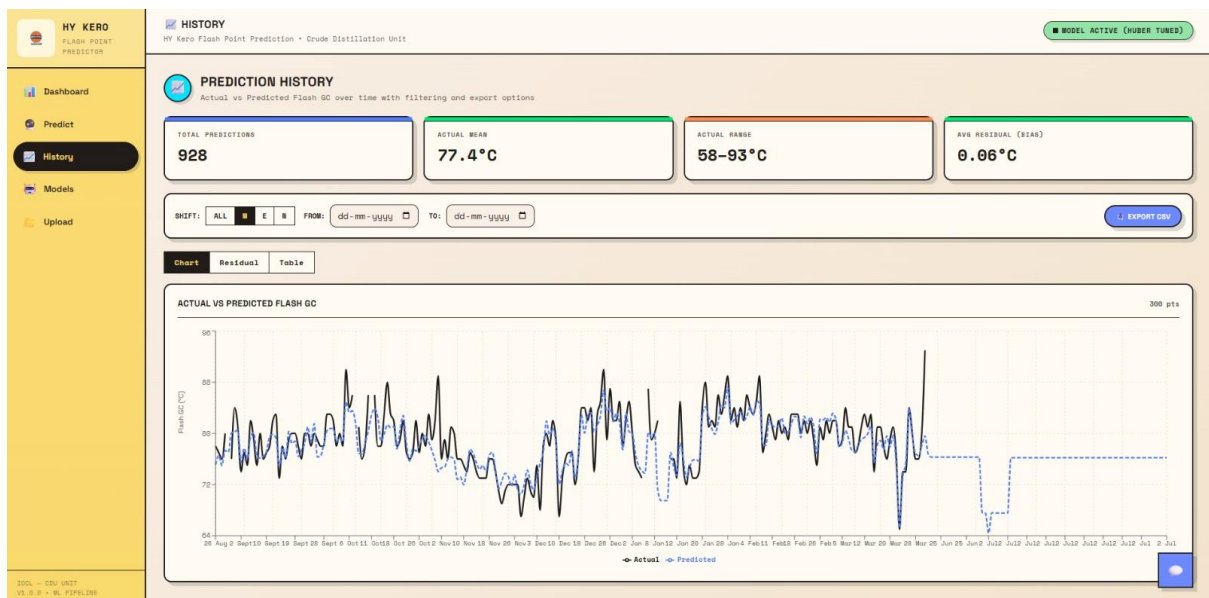
Predicted flash point is in spec 64.38°C at 2:49:47 pm.

100% - CPU UNIT

V1.0.0 - ML PIPELINE

Prediction History

Stores all previous predictions for future analysis.



Batch Prediction

Allows users to upload Excel or CSV files containing multiple process observations.

HY KERO
FLASH POINT
PREDICTOR

Dashboard

Predict

History

Models

Upload

100% - CSV URET
V1.0.0 - ML PIPELINE

UPLOAD

HY Kero Flash Point Prediction - Crude Distillation Unit

MODEL ACTIVE (HUBER TUNED)

BATCH PREDICTION UPLOAD

Upload a processed feature CSV or sensor Excel for batch Flash GC predictions

HOW IT WORKS

1. Upload your sensor data or feature CSV (matching features inside `data/processed/X_full.csv`)

2. Our FastAPI backend pre-processes the features and fills in any missing sensor tags

3. Predictions are computed instantly using the regularized Lasso/ElasticNet production model

4. Download the fully annotated CSV including 95% Confidence Interval limits

5. Need a template? Download the [sample CSV template](#) to see the required column structure

DROP CSV / XLSX HERE OR CLICK TO BROWSE

Supports .csv and .xlsx (Max 10MB)

Model Evaluation

Displays performance comparison among all machine learning algorithms.

HY KERO
FLASH POINT
PREDICTOR

Dashboard

Predict

History

Models

Upload

100% - CSV URET
V1.0.0 - ML PIPELINE

MODELS

HY Kero Flash Point Prediction - Crude Distillation Unit

MODEL ACTIVE (HUBER TUNED)

MODEL EVALUATION & OVERFITTING

Performance metrics and time-aware validation scores across all trained models

SELECTED PRODUCTION MODEL

Huber Tuned

Test RMSE: 0.22 | CV RMSE (5-Fold): 3.1 | Test R²: 0.6863

Comparison

Chart

Features

MODEL	TRAIN RMSE	TEST RMSE	CV RMSE (MEAN)	5-FOLD CV SCORES	TEST R ²	MAPE	OVERFIT?
Huber Tuned	2.698	0.22	3.1	[2.402, 3.116, 2.843, 4.008, 2.933]	0.6863	2.08%	☒ NO
Huber	2.617	0.244	4.671	[2.896, 3.202, 2.307, 3.487, 3.492]	0.6591	2.12%	☒ NO
Ridge Tuned	2.697	0.29	3.337	[2.898, 3.229, 3.937, 4.145, 3.191]	0.6572	2.09%	☒ NO
Ensemble	2.687	0.253	3.188	N/A	0.6563	2.08%	☒ NO
ElasticNet	2.688	0.264	3.285	[2.89, 3.18, 3.035, 4.028, 3.228]	0.65	2.09%	☒ NO
ElasticNet Tuned	2.688	0.281	3.105	[2.877, 3.146, 3.983, 3.988, 3.111]	0.6479	2.11%	☒ NO
BayesianRidge	2.599	0.266	4.678	[2.838, 3.202, 2.889, 3.647, 11.818]	0.6463	2.13%	☒ NO
Ridge	2.58	0.325	5.353	[2.664, 3.17, 2.465, 3.951, 14.932]	0.634	2.19%	☒ NO
Linear Regression	2.577	0.363	5.014	[2.849, 3.196, 2.484, 3.495, 10.113]	0.622	2.25%	☒ NO
Lasso	2.639	0.364	4.325	[2.956, 3.113, 2.476, 3.956, 9.218]	0.6215	2.24%	☒ NO
Lasso Tuned	2.698	0.393	3.118	[2.987, 3.118, 2.927, 3.871, 3.098]	0.6122	2.25%	☒ NO
XGBoost	2.717	0.56	3.905	[2.939, 3.989, 3.766, 4.657, 4.938]	0.1419	3.26%	☒ NO
LightGBM	2.728	0.698	3.96	[2.987, 3.992, 3.774, 4.676, 4.888]	0.0943	3.37%	☒ NO

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7.4 Advantages of Proposed System

- Real-time Flash Point prediction.
- Reduced laboratory dependency.
- Improved product quality.
- Better process optimization.
- Reduced energy consumption.
- Easy-to-use dashboard.
- Prediction history.
- Batch prediction support.
- Improved operational efficiency.

CHAPTER 8

DATA COLLECTION AND PREPROCESSING

8.1 Data Collection

Historical operational data were collected from the Crude Distillation Unit (CDU) at IOCL Mathura Refinery. The dataset contained process sensor readings and corresponding laboratory Flash Point values.

The collected data consisted of:

- Temperature sensors
- Pressure sensors
- Flow sensors
- Furnace outlet temperatures
- Stripping steam parameters
- Heavy Kerosene draw rates
- Laboratory Flash Point measurements

The data covered multiple operating conditions to ensure that the model could generalize effectively across different refinery scenarios.

8.2 Data Cleaning

Raw industrial datasets often contain missing values, duplicate records, inconsistent measurements, and outliers. Therefore, several preprocessing steps were applied before model training.

The preprocessing activities included:

- Removing duplicate records.
- Handling missing values.
- Removing invalid observations.
- Correcting inconsistent values.
- Synchronizing process variables with laboratory measurements.

These steps ensured the quality and reliability of the dataset.

8.3 Feature Engineering

Feature engineering was performed to improve the predictive capability of the machine learning models.

The following techniques were applied:

- Lag Features
- Rolling Mean
- Moving Average
- Ratio Features
- Time-based Features
- Derived Process Variables

Initially, 117 features were generated. Through feature selection techniques, the most relevant 30 features were retained for model training, reducing complexity while maintaining prediction accuracy.

8.4 Data Normalization

Since process variables have different numerical ranges, feature scaling was performed before training the regression models.

Normalization helped:

- Improve model convergence.
- Reduce bias due to varying feature scales.
- Increase prediction accuracy.
- Enhance training stability.

8.5 Data Flow Diagram

Historical Process Data



Data Cleaning



Missing Value Handling



Feature Engineering



Feature Selection



Data Scaling



Training Dataset

CHAPTER 9

MACHINE LEARNING MODEL DEVELOPMENT

9.1 Model Development

After preprocessing and feature engineering, several supervised machine learning regression algorithms were trained to predict the Heavy Kerosene Flash Point. Since the dataset consisted of historical refinery observations arranged chronologically, a **TimeSeriesSplit (gap = 2)** cross-validation strategy was adopted to prevent data leakage between training and testing datasets.

Initially, more than **117 engineered features** were generated from the raw process variables. These included:

- Temperature averages
- Flow averages
- Pressure averages
- Lag features
- Rolling averages
- Steam-to-Kerosene ratios
- Time-based features

To improve model performance and reduce complexity, feature selection techniques were applied, resulting in the final **30 most important process variables** being used for training.

9.2 Feature Selection Pipeline

The feature engineering process followed three sequential stages.

Stage	Technique	Outcome
Variance Threshold	Removed constant process variables	7 features removed
Correlation Analysis	Removed highly correlated sensors ($r > 0.92$)	17 features removed
Feature Importance	Selected highest weighted variables	Final 30 features

This process eliminated redundant information while preserving the most influential refinery process parameters.

9.3 Machine Learning Algorithms

The following regression algorithms were evaluated.

Model	Purpose
Linear Regression	Baseline Model
Ridge Regression	Regularized Linear Model
Lasso Regression	Feature Selection
ElasticNet	Combined Ridge & Lasso
Bayesian Ridge	Probabilistic Regression
Random Forest	Ensemble Learning
XGBoost	Gradient Boosting
LightGBM	Efficient Boosting
Huber Regression	Robust Industrial Prediction

9.4 Why Huber Regression was Selected

Although ensemble algorithms produced very low training errors, they failed to generalize under changing refinery operating conditions.

Huber Regression demonstrated superior industrial performance because it:

- Handles noisy laboratory measurements effectively.
- Is less sensitive to outliers.
- Performs well under process distribution shifts.
- Provides stable predictions during temperature variations.
- Prevents excessive influence of abnormal observations.

For these reasons, the **Huber Tuned Regression Model** was selected for deployment.

CHAPTER 10

MODEL PERFORMANCE EVALUATION

10.1 Performance Metrics

The developed models were evaluated using the following metrics:

- Root Mean Square Error (RMSE)
- Mean Absolute Error (MAE)
- Coefficient of Determination (R^2)
- Cross Validation RMSE

These metrics were selected because they accurately represent prediction quality for industrial regression problems.

10.2 Model Comparison

Model	CV RMSE	Test RMSE	Test R^2	Performance
XGBoost	2.145	4.882	-0.612	Overfitting
Random Forest	2.210	4.103	-0.134	Poor Generalization
Lasso Regression	2.190	2.301	0.6415	Good
Huber Tuned	2.091	2.220	0.6663	Best

10.3 Industrial Challenges

During evaluation, two major industrial challenges were identified.

Target Leakage

The previous model depended on laboratory Flash Point values from previous shifts.

This created several problems:

- Manual entry of previous laboratory values
- Failure when lab results were delayed

- Error propagation

The proposed feature engineering approach successfully removed this dependency.

Distribution Shift

During testing, the refinery operated under higher average temperatures (**+3.71°C** compared to the training period).

Tree-based algorithms such as Random Forest and XGBoost overfitted the historical data and failed to generalize.

Huber Regression maintained prediction accuracy because of its linear extrapolation capability.

10.4 Final Model Performance

The deployed model achieved:

Metric	Value
Test RMSE	2.22°C
Test R ²	0.6663
Cross Validation RMSE	2.091°C

Considering laboratory measurement uncertainty of approximately **1.5–2.0°C**, the developed model performs close to the theoretical accuracy limit.

CHAPTER 11

SYSTEM ARCHITECTURE AND WEB APPLICATION

The developed application follows a full-stack architecture integrating Machine Learning, Backend APIs, Database Management, and a React-based user interface.

11.1 Overall Architecture

Historical Sensor Data



Data Preprocessing



Feature Engineering



Huber Regression Model



FastAPI Backend



SQLite Database



React Dashboard

11.2 Backend

The backend was developed using **FastAPI**.

Major functionalities include:

- Loading trained model (.pkl)
- Loading feature scaler
- Prediction API
- Batch Prediction API
- History API
- AI Chatbot API

11.3 Database

SQLite was used for storing:

- Prediction History
- User Inputs
- Prediction Timestamp
- Confidence Interval
- Model Version

The database creates a complete audit trail for future analysis.

11.4 Frontend

The user interface was developed using **React.js**.

The dashboard consists of:

Dashboard

Displays

- Average Flash Point
- Prediction Statistics
- Trend Graphs

Prediction Module

Allows engineers to enter refinery sensor values and instantly obtain:

- Predicted Flash Point

- Confidence Interval
- Product Status

Batch Prediction

Allows uploading CSV/Excel files.

The application automatically predicts Flash Point for every observation.

History

Displays all historical predictions.

Features include:

- Search
- Filter
- Export

AI Assistant

The application integrates an AI-powered chatbot that helps operators understand process variables and retrieve prediction information using natural language.

CHAPTER 12

CONCLUSION AND INDUSTRIAL IMPACT

12.1 Conclusion

The internship project successfully demonstrated the practical implementation of Artificial Intelligence in refinery operations.

A real-time Heavy Kerosene Flash Point prediction system was developed using historical refinery sensor data and laboratory measurements.

After extensive experimentation, the **Huber Tuned Regression Model** achieved the best prediction accuracy with:

- **Test RMSE = 2.22°C**
- **$R^2 = 0.6663$**

The complete application was deployed as a full-stack web platform consisting of:

- FastAPI Backend
- React Dashboard
- SQLite Database
- Batch Prediction Module
- AI Assistant

The developed system enables refinery operators to predict Heavy Kerosene Flash Point in real time without waiting for laboratory analysis.

12.2 Industrial Impact

The proposed AI-based Soft Sensor provides several operational advantages:

- Eliminates dependence on previous laboratory Flash Point values.
- Reduces feedback time from **8 hours to real time**.
- Enables proactive process optimization.
- Improves Heavy Kerosene quality.
- Reduces energy losses caused by conservative operation.
- Provides a complete digital prediction history.

- Supports Industry 4.0 initiatives at IOCL.

12.3 Future Scope

The developed system can be further enhanced by:

- Integration with live DCS data.
- Automatic control recommendations.
- Deep Learning-based prediction models.
- Prediction of additional refinery quality parameters.
- Cloud deployment.
- Mobile application for refinery engineers.
- Integration with refinery digital twin platforms.